

Assignment 5

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Problem 1:

a) $v(t) = \mathcal{E}(t)$, $v(t) = 10 \cos(100t)$, $R = 100 \frac{\text{A} \cdot \text{turns}}{\text{Wb}}$, $N = 4$

$$\mathcal{E}(t) = N \frac{d\phi}{dt} \Rightarrow \phi(t) = \int \frac{v(t)}{N} dt$$

$$\phi(t) = \frac{1}{4} \cdot \int 10 \cos(100t) dt = \frac{1}{4} \cdot \frac{10}{100} \sin(100t) + C$$

$$C = \phi(0) = 0 \Rightarrow \phi(t) = \frac{1}{40} \sin(100t)$$

$$NI = R\phi \Rightarrow I = \frac{R\phi}{N} = \frac{100}{4} \cdot \frac{1}{40} \sin(100t) = \frac{5}{8} \sin(100t)$$

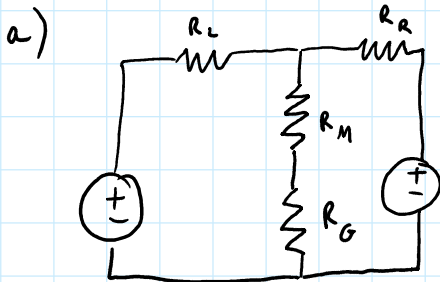
b) $V_L = N \frac{d\phi}{dt}$, $Ni = R\phi$, $V_L = L \frac{di_L}{dt}$

$$N \frac{d\phi}{dt} = L \frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{R}{N} \frac{d\phi}{dt} \Rightarrow N \frac{d\phi}{dt} = \frac{RL}{N} \frac{d\phi}{dt}$$

$$L = \frac{N^2}{R}$$

Problem 2:



b) $F = \phi R = Ni \Rightarrow i = \frac{\phi R}{N}$

$$R = R_L + R_R + R_m + R_G \cdot 2$$

$$R_L = R_R = \frac{L_L}{\mu_{\text{iron}} \cdot A_{\text{core}}} = \frac{(0,06 \cdot 2 + 0,08) \text{ m}}{5000 \cdot 4\pi \cdot 10^{-7} \cdot (0,02)_{\text{m}}^2} = 79577$$

$$R_m = \frac{(0,08 - 0,001) \text{ m}}{5000 \cdot 4\pi \cdot 10^{-7} \cdot (0,02)_{\text{m}}^2} = 31433$$

$$R_G = \frac{0,001 \text{ m}}{4\pi \cdot 10^{-7} \cdot (0,02 \text{ m})^2} = 1989436$$

$$R = 4169459$$

$$\Phi = B_{\text{gap}} \cdot A_{\text{gap}} = 0,25 \text{ T} \cdot (0,02 \text{ m})^2 = 10^{-4} \text{ wb}$$

$$i = \frac{\Phi R}{N} = \frac{10^{-4} \cdot 4169459}{200} = 2,08 \text{ A}$$

Problem 3:

$$B(r) = \frac{\mu_0 I}{2\pi r}$$

$$\Phi = \int_{r_0}^{r_0+b} B(r) h dr = \int_{r_0}^{r_0+b} \frac{\mu_0}{2\pi r} h dr = \frac{\mu_0 I h}{2\pi} \ln\left(\frac{r_0+b}{r_0}\right)$$

$$M = \frac{\Phi}{I} = \frac{\mu_0 h}{2\pi} \ln\left(\frac{r_0+b}{r_0}\right)$$

Problem 4:

$$a) S_1 = V_1 I_1^* = V_2 I_2^* = S_2$$

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} \Rightarrow V_1 I_1^* = V_2 I_2^*$$

$$\frac{I_2^*}{I_1^*} = \frac{N_1}{N_2} \Rightarrow \frac{I_2}{I_1} = \frac{N_1}{N_2}$$

$$\Rightarrow I_1 N_1 = I_2 N_2$$

$$b) \frac{V_S}{V_L} = \frac{N_1}{N_2} \Rightarrow V_L = \frac{N_2}{N_1} V_S = \frac{1}{50} \cdot 1000 e^{j0^\circ} = 20 e^{j0^\circ} \text{ V}_{\text{rms}}$$

$$Z_L = 0,5 + j0,5 \Omega$$

$$I_2 = \frac{V_L}{Z_L} = \frac{20 e^{j0^\circ}}{0,707 e^{j45^\circ}} = 28,28 e^{-j45^\circ} \text{ A}$$

$$\frac{I_1}{I_2} = \frac{N_2}{N_1} = \frac{1}{50}$$

$$I_1 = \frac{1}{50} I_2 = \frac{1}{50} (28,28 e^{-j45^\circ}) = 0,5656 e^{-j45^\circ} A$$

c) Load Z_L :

$$S_L = V_L I_2^* = 20 e^{j0^\circ} \cdot 28,28 e^{j45^\circ} \approx 565,6 \angle 45^\circ VA$$

Rect. form: $400 + j400 VA$

Load consumes 400W of active power
and 400var of reactive.

Source V_s :

$$S_s = V_s I_1^* = 1000 e^{j0^\circ} \cdot 0,5656 e^{j45^\circ} = 400 + j400 VA$$

→ Delivers 400W and 400var

Problem 5:

$$\frac{|V_{ab} \cdot 20|^2}{Z_L} = \frac{|V_{ab}|^2}{Z_{ab}}$$

$$\Rightarrow \frac{25}{4} Z_L = Z_{ab} \Rightarrow Z_{ab} = 1250 e^{-j\frac{\pi}{4}}$$